Engineering Mathematics - I · Unit_03 · Unit 3

DRAFT — UE26MA141A · Unit_03 · Unit 3 · Lagranges Linear equation continued

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Lagranges Linear equation continued

1 Lagranges Linear equation continued

Study focus: Lagranges Linear equation_continued. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000046 · EV0000047

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In this session, we will solve problems on Lagrange's linear equation.

Key details

- Q equals R of XYZ.
- After the selection of PQR, we write the acceler equations.
- That is dx divided by y square z over x, dy by x and dz by y square.
- to be x times dx divided by y square z equals dy by x z and dz by y square.
- So we have to think which method is appropriate after writing such a equation.
- Should we go for grouping or should we go for multipliers or can go for both.
- When we observe these actual equations, we can solve this problem by method of grouping because we can separate the variables and we can perform integration.
- we'll get x square dx and y square dy.
- And this is same as x power 3 minus y power 3 equals 3c, which is same as some constant a.
- So the first solution for this problem is going to be x power 3 minus y power 3 equals a.
- Again, when we'll observe this exer equation, we can go for grouping again.
- So if you take a look at this, if you take first and the last ratios.
- We see that x dx divided by y square z equals dy dz divided by y square.
- After cancellation of y square and this y square, that leads to x into dx equals z time dz.
- Again variables have been separated.
- As from theory, once we find out these two linearly independent solutions, we write the general solution to be phi of u, comma v 0 so which is same as phi of x power 3 minus y power 3 and x square minus z square equals 0.
- So in this case we took the first and second fractions and first and last fractions.
- If it is possible to take second and third, it is also allowed.

Example or application from the lecture

- On noting these idea, let us solve problems on Lagrangian's linear equation.
- Let us look at yz square divided by x times small p plus x q equals y square.
- In order to make simplification easy, let us simplify the first term.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000046 · EV0000047 · EV0000048 · +1 source units in pack.json

Core explanation · 05:00–09:59

Let us take a look at the second problem.

Key details

- It is yzp minus xzq equals xy.
- Dy divided by minus x and dz by xy.
- Again we can easily decide we can go for method of grouping in order to find both the solutions.
- So from the first and second fractions, we get dx by y equals dy by minus x.
- After integration that becomes x square by 2 plus y square by 2 equal to some constant c1.
- That is same as x square plus y square equal to 2 c1.
- So this is the first solution for the problem.
- So to find the second solution again we can go for the first and third fractions that gives.
- Dx by Y Z and Dz by XY After cancellation of this y and this y variables are separated.
- This is same as x into dx and z into dz.
- So this is how we can apply method of grouping for a problem to solve Lagrangius linear equation.
- This is going to be capital Q and this is going to be R.
- So while looking at the actual equation, if you go for first and two fractions after cancellation of Z square plus XY and this Z square plus XY we get dx by x equals dy by minus y.
- And you can cancel this z also.
- If you take a look at right hand side, unless we solve for this xy, we cannot say this is purely a function of z.
- In order to do that, we can use the previous answer.
- that is xy equal to c1 so when you replace x y by c1 so we get this there so x power 3 dx equals z power 3 plus c1 z time dz now you can see the left hand side we have a x and dx on the right hand side we have z and dz so now you can integrate.

Example or application from the lecture

- Let us take out another problem Z of Z square plus XY times p x minus q, everything is a product.
- So now how do you find out second solution that is b equals to b let us take out the first and third fractions so we get the x power 3 dx and z power 3 plus xy z tim z.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000050 · EV0000051 · EV0000052 · +1 source units in pack.json

Core explanation · 10:00–14:59

This is z power 4 by 4, c1 is constant, z becomes z square by 2 plus another constant, namely c2.

Key details

- So these are the problems on Lagrangius linear equation and to find out the solution we use the method of grouping.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000054 · EV0000055 · EV0000056 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Lagranges Linear equation_continued transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/003_Lagranges_Linear_equation_continued__dba217df524461e2.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Lagranges Linear equation_continued transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/003_Lagranges_Linear_equation_continued__dba217df524461e2.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Lagranges Linear equation_continued transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/003_Lagranges_Linear_equation_continued__dba217df524461e2.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9989; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Lagranges Linear equation_continued transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_03/AV_Summaries/003_Lagranges_Linear_equation_continued__dba217df524461e2.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
2. 1 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
3. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
4. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
5. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
6. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
7. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00018	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0040. Sidecar variants compare...
SRC00019	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0040. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00020	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0040. Sidecar variants compare...
SRC00021	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0038 to s0040. Sidecar variants compare...

C Revision and retrieval practice